

Customer Complaint Triage & Escalation Classifier

Transformer-based routing for CFPB financial complaint narratives

Final Presentation

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Problem Statement

Manual complaint triage is slow

- Free-text narratives are long, noisy, and inconsistent.
- Routing errors delay resolution and increase specialist workload.
- Urgent complaints may be missed without prioritization.

ML system goal

- Predict the correct operational routing category.
- Flag complaints that need urgent human review.
- Turn unstructured text into measurable workflow decisions.

Core idea: classify the narrative, then route or escalate the ticket.

Dataset: CFPB Consumer Complaint Database

Source

CFPB

Official public complaint database

Input text

Narrative

Consumer-submitted text

Subset used

8,328

Train + validation + test rows

Test set

1,528

Held-out evaluation examples

Field group	Examples used in this project
Narrative	Consumer complaint narrative
Product metadata	Product, sub-product, issue, sub-issue
Company context	Company, company public response, company response to consumer
Process signals	Date received, submitted via, timely response, complaint ID

CFPB database contains complaints about consumer financial products and services.

Database documents fields such as Product, Issue, Sub-issue, Consumer complaint narrative...

Source: <https://www.consumerfinance.gov/data-research/consumer-complaints>

Two supervised learning tasks

Input

Consumer complaint narrative

Example signal: “I did not authorize this account,” “payment was applied incorrectly,” “documents were never processed.”

Task A: triage routing

- 6 custom operational categories
- Multi-class classification
- Goal: send complaint to the correct review team

CUSTOM LABELS

Task B: escalation

- Binary classification
- Escalate vs do not escalate
- Goal: prioritize urgent human review

COST-SENSITIVE

Data extraction and reproducible pipeline

What to extract:

- complaint ID
- narrative text
- product
- sub-product
- issue
- sub-issue
- company
- company public response
- timely response
- date received

Showing 10,856,593 matches out of 14,380,839 total complaints

[Export data](#) [Print](#)

[Trends](#)

List

[Map](#)

Show

Sort

25 results

Newest to oldest

Read

Only complaints with narratives

All complaints

Repository workflow

- Place CSV at data/raw/complaints.csv
- Profile and clean raw complaint data
- Bootstrap weak labels with rules
- Manually review the annotation queue
- Create stratified train/val/test splits
- Train baseline + transformer models
- Evaluate and run error analysis

Manual labeling : operational categories

Triage label	Meaning in the workflow
fraud_identity_risk	Fraud, scam, identity theft, unauthorized accounts
billing_payment_dispute	Charges, refunds, payments, billing disputes
account_access_management	Opening, closing, login, account access or management
documentation_processing_verification	Documents, processing delays, verification, investigations
customer_service_failure	Poor support, unresolved contact, broken service process
legal_compliance_regulatory	Legal threats, regulation, credit reporting violations, compliance risk

Labeling process

- Weak-label rules generated candidate labels.
- Manual review queue supports correction.
- Annotation guidelines define tie-break rules.

Escalation label

- Escalate: urgent fraud, legal, regulatory, severe harm, or high-risk complaint.
- Do not escalate: standard operational routing without urgent specialist review.

Preprocessing:

Cleaning decisions

- Kept realistic complaint wording for transformers.
- Dropped rows without public narratives.
- Removed empty or extremely short narratives.
- Normalized whitespace and duplicated records.
- Stored raw and cleaned text for reproducibility.

Why not over-clean?

- Transformer tokenizers expect natural text.
- Aggressive cleaning can remove domain-specific signals.
- Complaint wording itself contains urgency, frustration, and risk clues.

Train split

5600

Used for model learning

Validation split

1200

Model selection + training curves

Test split

1528

Final held-out evaluation

Literature review: zero-shot LLM benchmark on CFPB complaints

Reference study

- Roumeliotis et al. evaluated zero-shot complaint classification using leading LLMs
- Models included Claude 3.5 Sonnet, Claude 3.5 Haiku, GPT-4o, GPT-4o-mini, and DeepSeek Chat
- The task used five broad financial product classes, not custom operational routing labels.

Model	Acc.	Precision	Recall	F1	Cost
Claude 3.5 Sonnet	0.763	0.797	0.763	0.762	\$0.64
DeepSeek Chat	0.737	0.752	0.737	0.737	\$0.01
GPT-4o	0.736	0.755	0.736	0.736	\$0.45
Claude 3.5 Haiku	0.721	0.729	0.721	0.723	\$0.17
GPT-4o-mini	0.691	0.703	0.691	0.693	\$0.03

Key takeaway: strong zero-shot LLMs work well, but cost and task design matter.

How this project differs from the literature

Dimension	Reference study	This project
Learning setup	Zero-shot LLM prompting	Fine-tuned supervised classifiers
Labels	5 broad product categories	6 operational triage labels + escalation
Goal	Benchmark model accuracy	Support business workflow decisions
Cost profile	API cost per inference	Local/controlled model inference after training
Operational output	Complaint product class	Routing category + urgent review flag

Important comparison rule

The literature is useful for context, but the tasks, labels, and deployment assumptions are different.

Academic contribution

This project shifts the problem from broad product classification to operational decision support: route the complaint and prioritize urgent review.

Model selection and Fine-Tuning

Baseline

- TF-IDF text features
- Logistic Regression classifier
- Fast, interpretable benchmark
- Useful to prove transformers add value

DistilBERT

- Transformer sequence classifier
- Smaller and faster than BERT-style models
- Fine-tuned on custom labels
- Chosen as practical proposed model

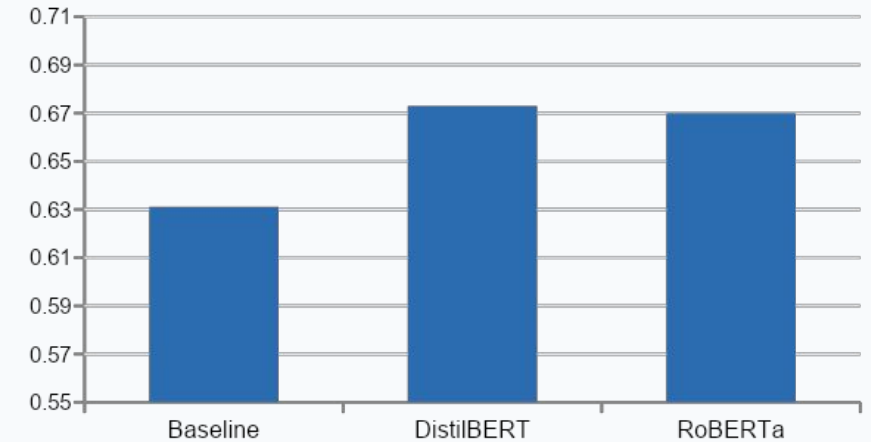
RoBERTa

- Stronger transformer baseline
- Good comparison model for classification
- Same train/validation/test split
- Used to test whether larger contextual model improves results

Training used Hugging Face Transformers with GPU acceleration on an NVIDIA RTX 3060 Ti

Triage classification results:

Model	Accuracy	Macro F1	Weighted F1	Interpretation
TF-IDF + Logistic Regression	0.664	0.631	0.666	Strong traditional baseline
DistilBERT	0.700	0.673	0.704	Best balanced model
RoBERTa	0.700	0.670	0.702	Best fraud recall



Best accuracy

0.700

DistilBERT / RoBERTa tie

Best macro F1

0.673

DistilBERT

Best weighted F1

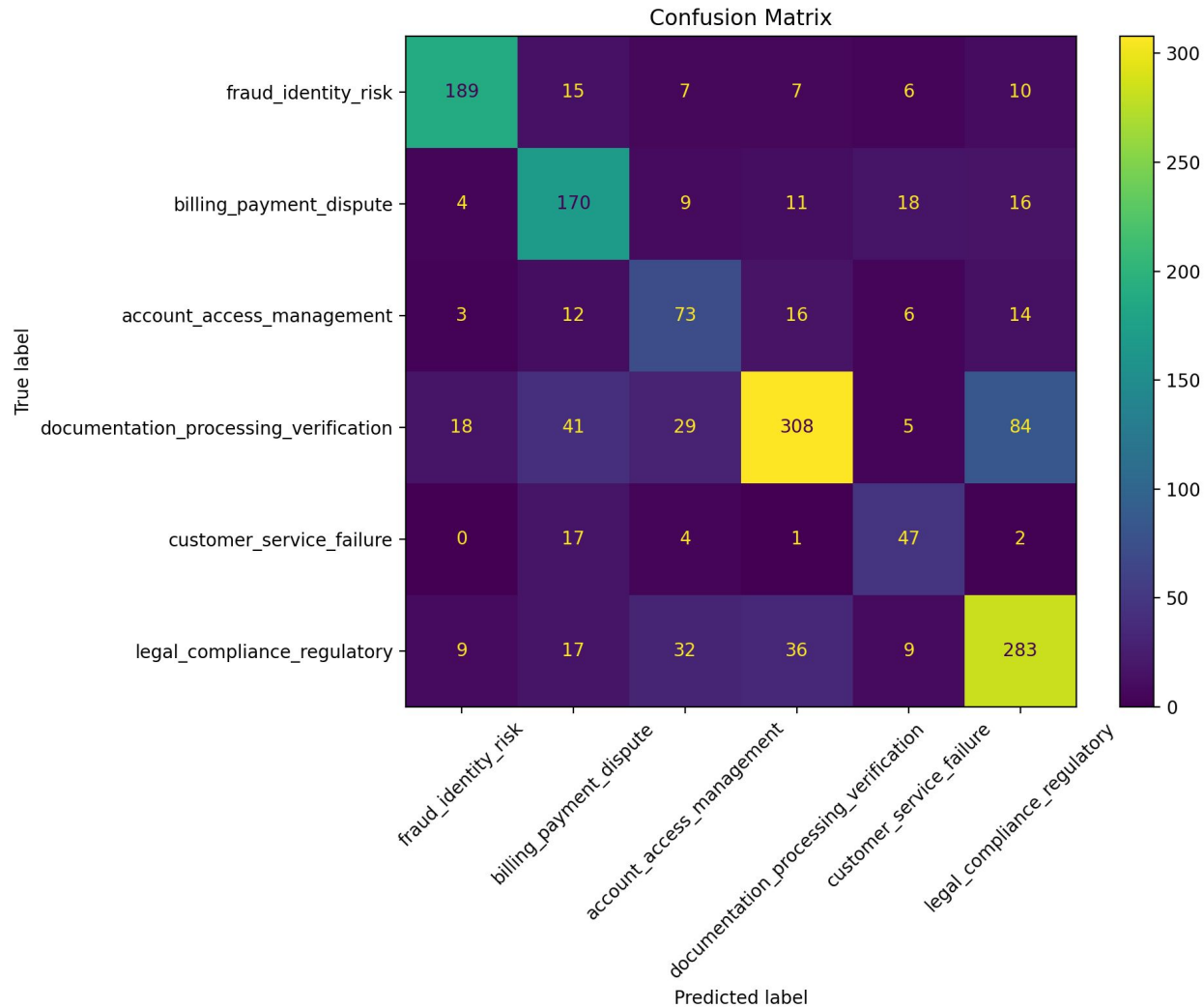
0.704

DistilBERT

Main conclusion

DistilBERT is the strongest overall triage model because it ties RoBERTa in accuracy while achieving slightly better macro and weighted F1.

Triage error analysis: where the model struggles



Strongest class

Fraud / identity risk reached the best F1 score under DistilBERT: 0.827

Hardest class

Account access management remained weakest because it overlaps with billing, fraud, documentation, and service failures

Main confusion

Documentation / processing and legal / compliance are often confused because both use investigation, verification, and regulatory language.

Escalation results: prioritizing urgent human review

Model	Accuracy	Macro F1	Precision	Recall	F1
TF-IDF + Logistic Regression	0.786	0.758	0.590	0.788	0.675
DistilBERT	0.817	0.781	0.656	0.735	0.693

Fewer false positives

166

DistilBERT vs. 236 baseline

Higher precision

0.656

Cleaner escalation queue

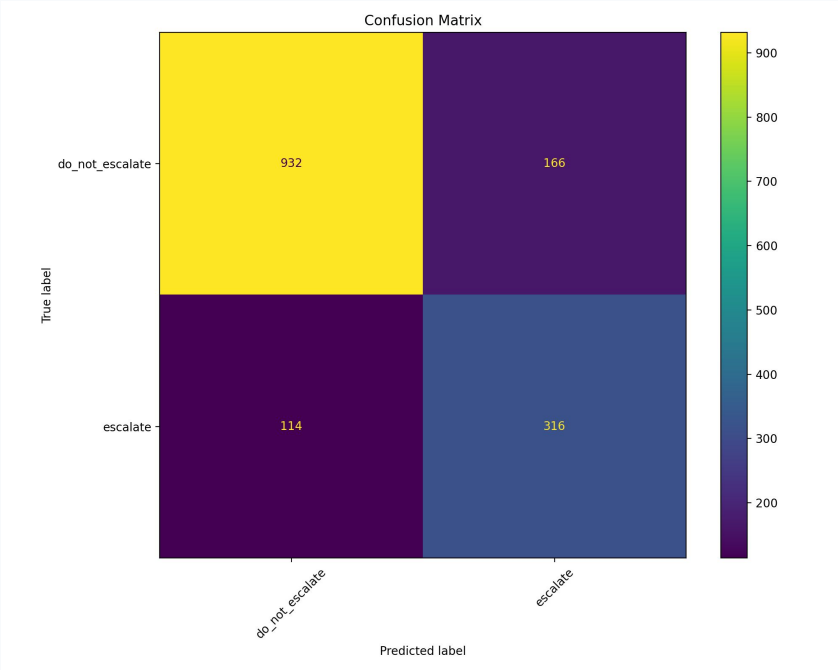
Higher macro F1

0.781

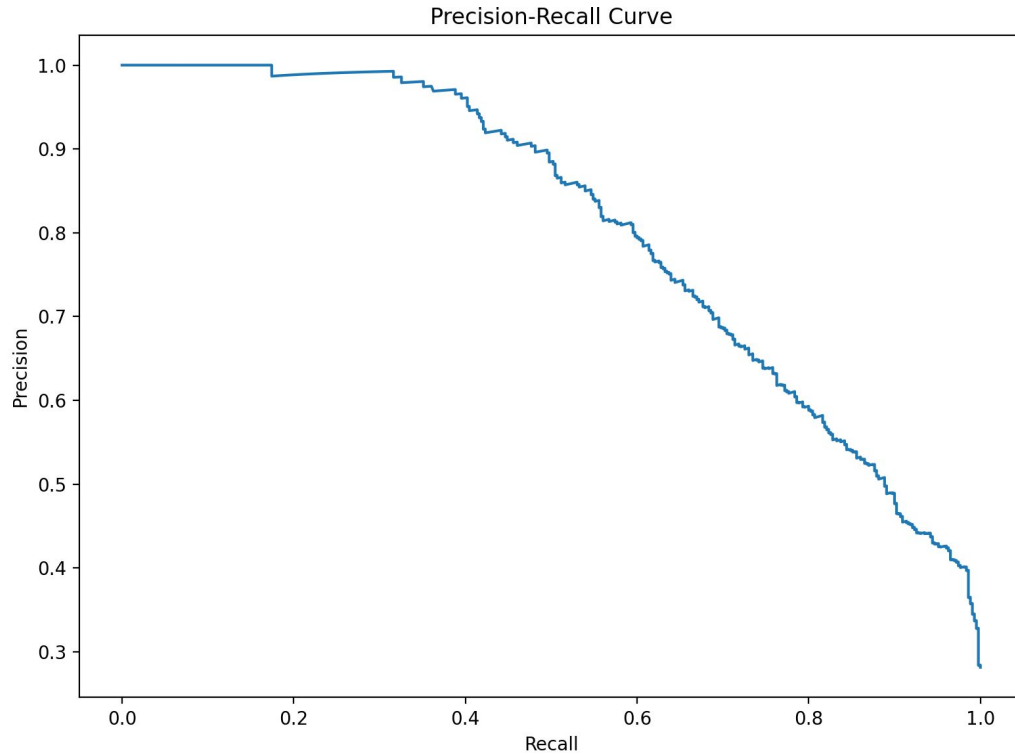
Best overall escalation model

Tradeoff

The baseline catches more urgent cases, but DistilBERT reduces unnecessary escalations and improves overall balanced performance.

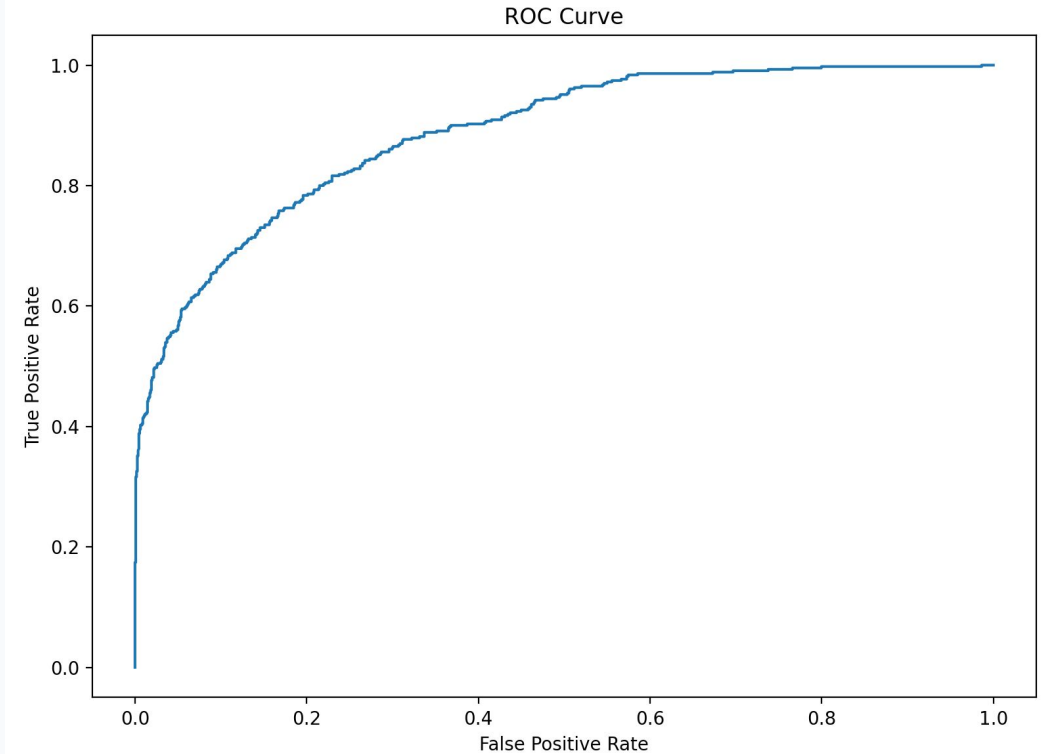


Threshold tuning matters for escalation



Precision-recall curve

Best for imbalanced escalation because it shows the cost of catching more urgent cases.



ROC curve

Shows ranking ability but can be optimistic when one class is larger.

Proposed model: DistilBERT

Why DistilBERT?

- Best triage macro F1 and weighted F1.
- Best escalation accuracy and macro F1.
- Lower complexity than larger transformer models.
- Good balance of quality, speed, and deployability.

Decision factor	Preferred model	Reason
General triage	DistilBERT	Best balanced metrics
Fraud recall	RoBERTa	Highest fraud/identity recall
Escalation queue quality	DistilBERT	Higher precision and fewer false positives
Maximum escalation recall	Baseline	Catches more urgent cases at default threshold
Production candidate	DistilBERT	Best overall balance

Deployment

DistilBERT can be used as the default mode.

Limitations and future work

Current limitations

- Labels were bootstrapped and need deeper manual validation.
- Classes are imbalanced, especially customer service failure.
- Some labels are semantically overlapping by design.
- The selected subset is smaller than the full CFPB database. (14.5 Million)
- Default threshold may not reflect business risk tolerance.

Future work

- Review more labels manually and measure annotator agreement.
- Train on a larger CFPB subset or full dataset with chunked loading.
- Tune escalation threshold
- Add calibration, confidence scores
- Test additional models such as MiniLM or DeBERTa.

Conclusion

This project moves complaint NLP beyond sentiment analysis into operational decision support.

What was built

- A reproducible Python repo
- Custom triage and escalation labels
- Baseline + transformer models

What was learned

- Transformers improved triage results
- DistilBERT best balanced model
- Threshold tuning is needed for escalation

Why it matters

- Faster routing
- Cleaner escalation queue
- Better prioritization for compliance risk

Proposed model: DistilBERT fine-tuned on custom complaint-routing labels

References

- Consumer Financial Protection Bureau. Consumer Complaint Database
- Hugging Face Transformers documentation.
- Roumeliotis, K. I., Tselikas, N. D., & Nasiopoulos, D. K. “DeepSeek and GPT Fall Behind: Claude Leads in Zero-Shot Consumer Complaints Classification.” Preprints/Electronics, 2025.